IMPase: evidence before behavior

Michael Emanuel Glover / Virtual Cell research extension

Thirty traceable figure markers from recombinant bovine enzyme assays support conditional rate-law comparison. Seven Mg points and fifteen lithium points are used for fitting; three Mg points and five points at a held-out lithium concentration are evaluated afterward.

ins1p_magnesium: single_site_inhibition
Held-out native-rate RMSE: 0.0034708 Delta A360/min.

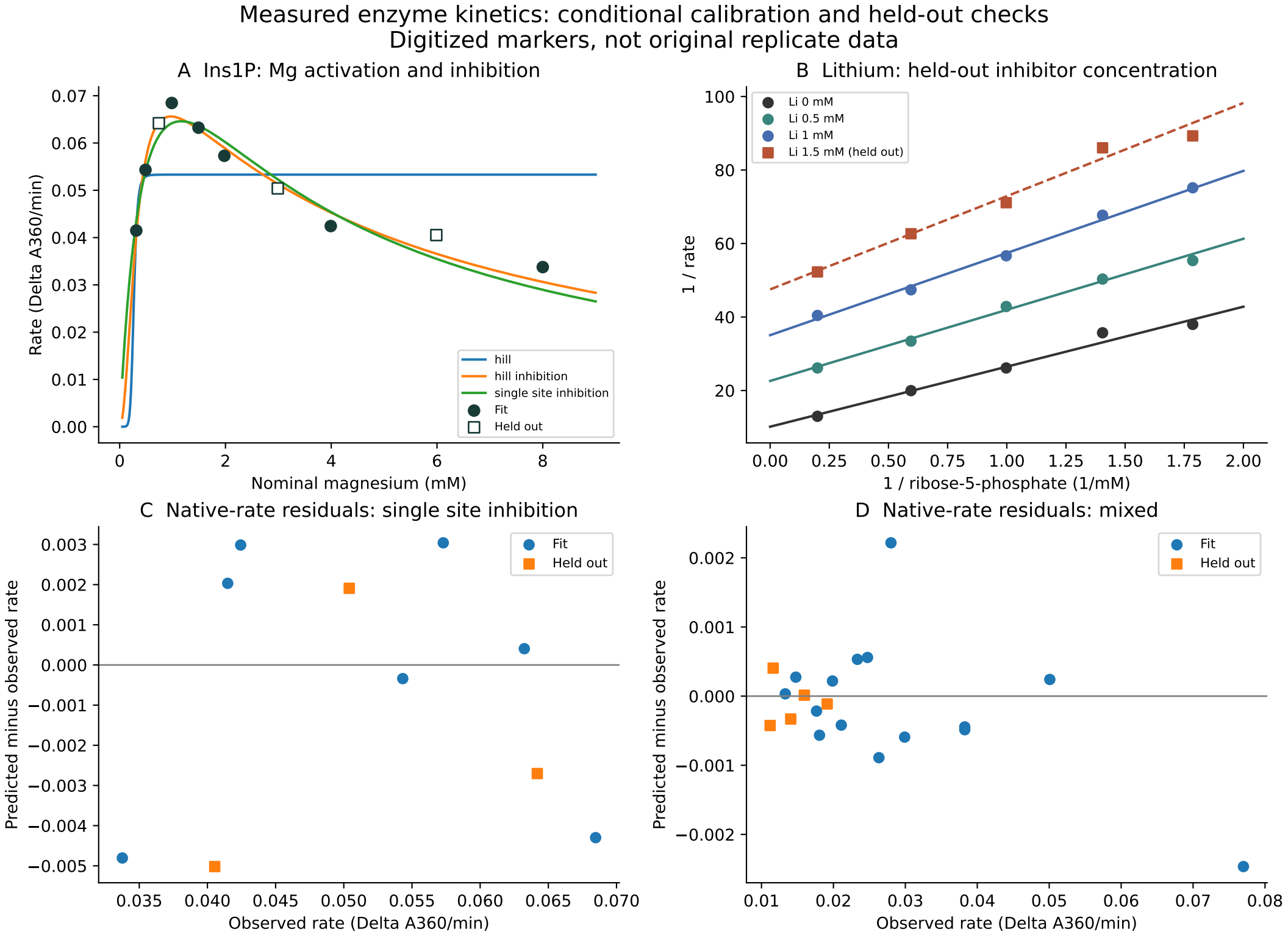
ribose_lithium: mixed
Held-out native-rate RMSE: 0.00030555 Delta A360/min.

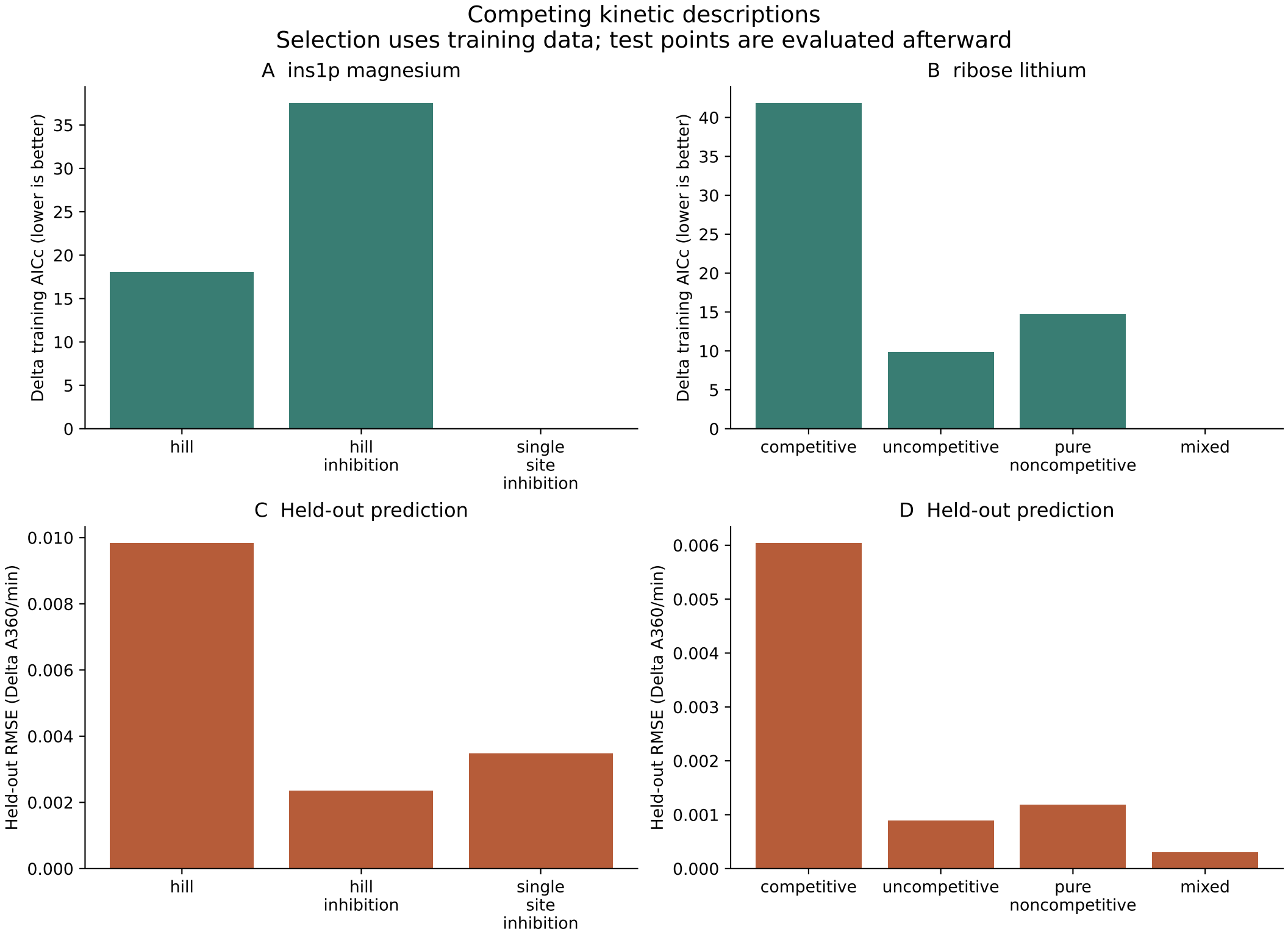
These are within-figure retrospective checks, not an independent replication. Coordinate perturbations quantify digitization sensitivity, not experimental standard errors.

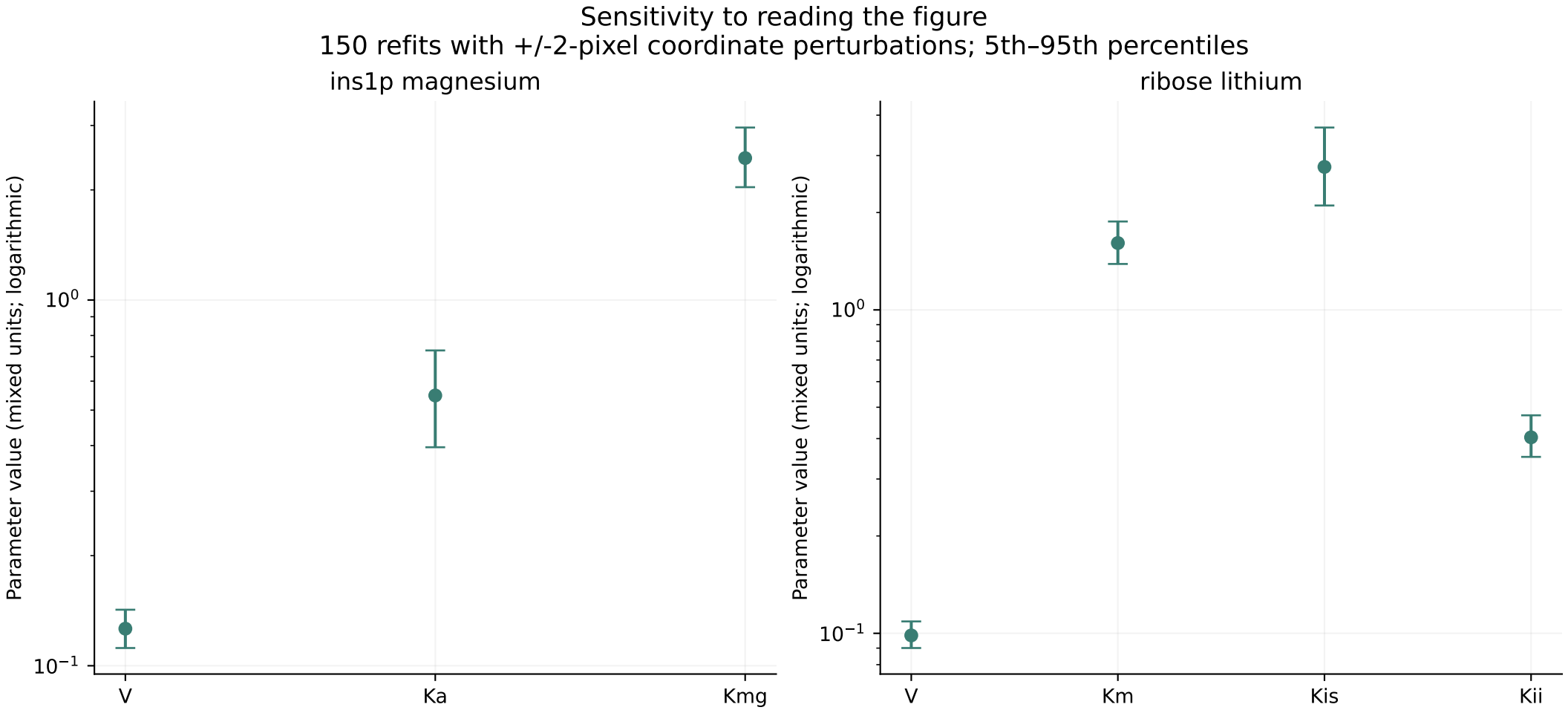
The assay enzyme is bovine, not human. Rates remain in their measured optical units. No fitted rate is converted into mood, neuronal firing or a universal physiological parameter.

Source: Strasser, Pelton and Ganzhorn (1995), Biochemical Journal 307:585–593, doi:10.1042/bj3070585.

Read REPORT.md for full methods, interpretation, parameter units, identifiability, source conditions, prediction tables and the next experiment.


Measured enzyme kinetics: conditional calibration and held-out checks
Digitized markers, not original replicate data
A Ins1P: Mg activation and inhibition
Rate (Delta A360/min)
Nominal magnesium (mM)
hill
hill inhibition
single site inhibition
Fit
Held out
B Lithium: held-out inhibitor concentration
1 / rate
1 / ribose-5-phosphate (1/mM)
Li 0 mM
Li 0.5 mM
Li 1 mM
Li 1.5 mM (held out)
C Native-rate residuals: single site inhibition
Predicted minus observed rate
Observed rate (Delta A360/min)
Fit
Held out
D Native-rate residuals: mixed
Predicted minus observed rate
Observed rate (Delta A360/min)
Fit
Held out


Competing kinetic descriptions
Selection uses training data; test points are evaluated afterward
A ins1p magnesium
B ribose lithium
C Held-out prediction
D Held-out prediction
Delta training AICc (lower is better)
Held-out RMSE (Delta A360/min)
hill
hill inhibition
single site inhibition
competitive
uncompetitive
pure noncompetitive
mixed


Sensitivity to reading the figure
150 refits with +/-2-pixel coordinate perturbations; 5th–95th percentiles
ins1p magnesium
ribose lithium
Parameter value (mixed units; logarithmic)
10^{0}
10^{-1}
V
Ka
Kmg
Km
Kis
Kii